Aadit Kannan

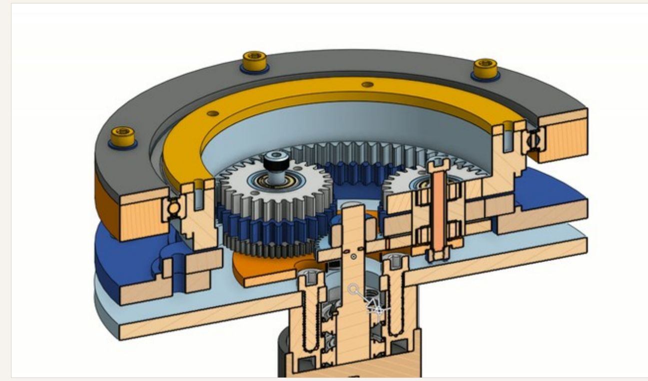
Robotics hardware, compact actuation, HV systems, PCB instrumentation, CAD/DFM

Mechanical engineering and EECS student at UC Berkeley. This packet reframes selected aaditkannan.com project work around practical robotics hardware: packaging constraints, actuator prototypes, electromechanical integration, manufacturing drawings, bench validation, and design-build-test iteration.

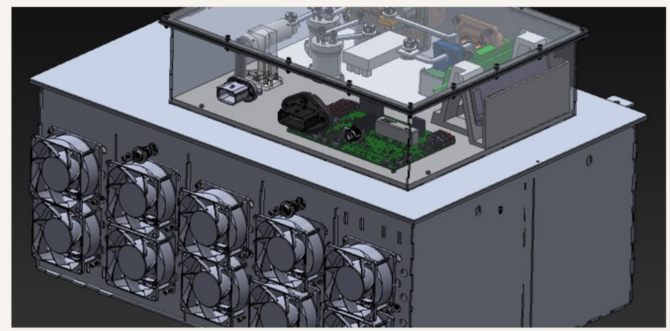
PROJECT INDEX

LINKS

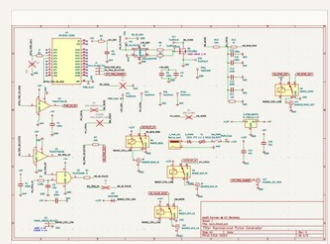
aaditkannan.com **aaditkannan.com/projects**



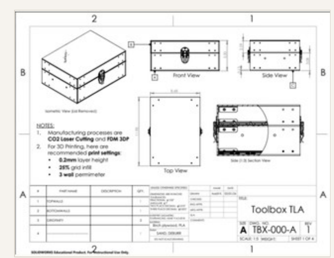
Wolfrom compound planetary section



588 V accumulator and HV electronics attic CAD

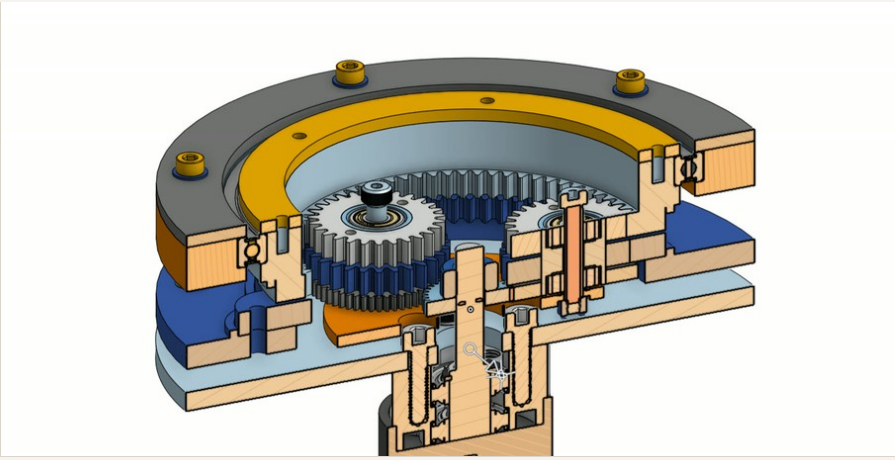


Nanosecond pulse generator schematic

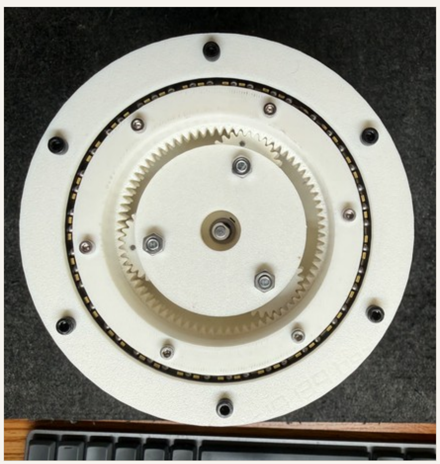


Toolbox drawing package

01 / Compact Wolfrom Compound Planetary Actuator



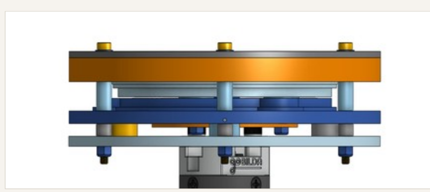
Section view: driven sun, compound planets, fixed ring, output ring, bearing support.



Printed prototype used for gear mesh and stackup checks.

Bench assembly / fit-up workflow.

Output-side bolt pattern and ring geometry.



Axial stack height and housing interfaces.

PROJECT SNAPSHOT

PROBLEM	Compact high-ratio joint actuation without bulky stacked stages
ROLE	Mechanical architecture, CAD, gearbox stackup, prototype planning
ARCHITECTURE	Driven sun + compound planets + fixed ring + output internal ring
TARGET	Approximately 50:1 reduction in compact axial package
STATUS	Printed prototype iteration; no production torque/efficiency claims
TOOLS	Onshape, CAD, gearbox design, bearing/fastener packaging, FDM/SLA

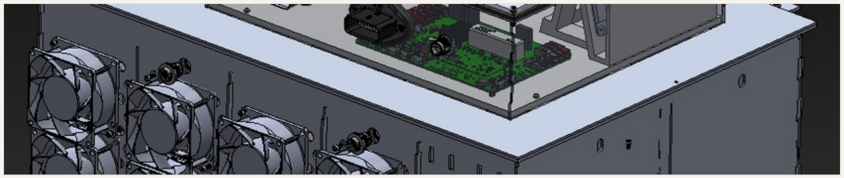
HIGH-SIGNAL ENGINEERING WORK

- Modeled sun gear, compound planet stacks, fixed ring, output ring, carrier, bearings, shafts, output hub, and housing interfaces.
- Iterated output ring splitting, bolt access, bearing support, axial spacing, and printability around an actual assembly sequence.
- Used prototypes to evaluate gear meshing, packaging, mechanical fit, and where a later metal/SLA hybrid version needs stronger interfaces.
- Main constraints: gear clearance, output ring stiffness, shaft alignment, bearing support, fastener access, printability, and manufacturable transition path.

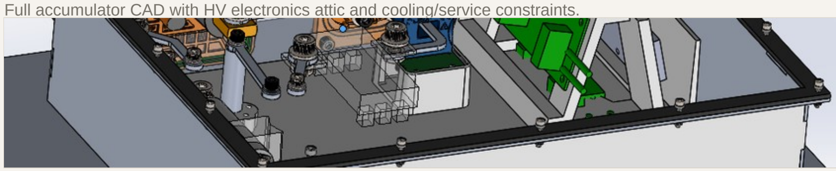
02-03 / Electromechanical Systems + Lab Instrumentation

Formula Electric Berkeley - 588 V Accumulator / HV Electronics Attic

HV battery packaging, service access, cooling, electronics integration



Full accumulator CAD with HV electronics attic and cooling/service constraints.



Electronics attic and enclosure-level packaging detail.

SNAPSHOT

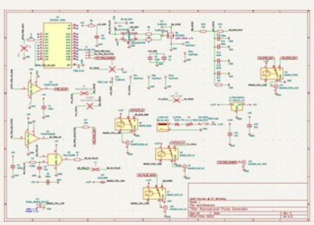
PROBLEM	Dense safety-critical EV hardware package
ROLE	Custom accumulator and electronics attic packaging
FOCUS	HV isolation, airflow, board/connector access, serviceability
TOOLS	SolidWorks, electromechanical packaging, DFM

ENGINEERING WORK

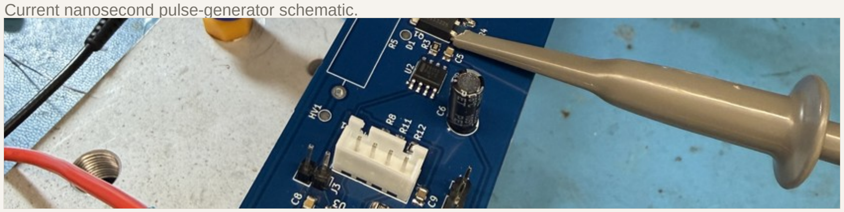
- Packaged enclosure, electronics attic, boards, fans, fasteners, and service interfaces in one constrained system.
- Balanced structural packaging, high-voltage routing/isolation, inspection access, cooling path, and competition vehicle constraints.
- Designed around maintainability rather than treating the accumulator as a static box.

Ramesh Lab - Nanosecond / Microsecond Pulse Generator PCB

BFO thin-film electrical characterization, triggering, bench validation



Current nanosecond pulse-generator schematic.



V1 microsecond board during oscilloscope probing.

SNAPSHOT

PROBLEM	Controlled pulses for BiFeO3 switching/transport tests
ROLE	Schematic/layout direction and validation workflow
FOCUS	GaN switching, trigger/sync paths, grounded coax/BNC-style interfaces
TOOLS	KiCad, LTspice, Python, LabVIEW, JupyterLab, oscilloscope

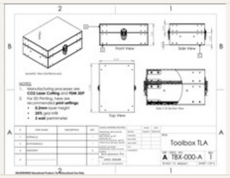
ENGINEERING WORK

- Design supports tests around the current 30 V regime before smaller-amplitude sweeps.
- Separates switching, leakage, displacement current, and transport response through repeatable triggering and waveform capture.
- V1 microsecond board informed the need for tighter layout, cleaner sync, and better grounded lab interfaces.

04-05 / Manufacturing + Robot Mechanisms

Custom Toolbox Design & Manufacturing

CAD to drawings to tolerancing to fabrication to inspection



Assembly drawing defining lid, latch, hinge, body, and insert interfaces.

Prototype fit-up photo; final-version photos unavailable.

SNAPSHOT

PROBLEM	Practice real mechanical part communication and fit-up control
ROLE	CAD assembly, drawings, GD&T-style tolerancing, fabrication planning
FOCUS	Datums, hole patterns, fastener alignment, lid/latch/hinge fit
TOOLS	SolidWorks, ASME Y14.5-style GD&T, laser cutting, FDM, calipers

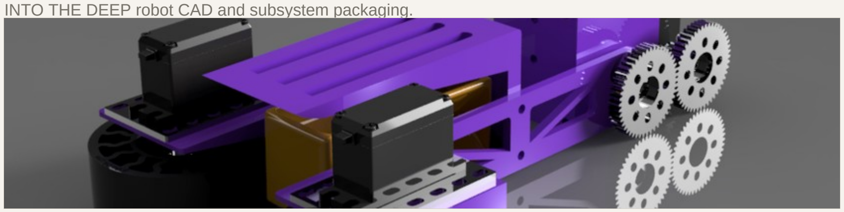
ENGINEERING WORK

- Generated production-style part and assembly drawings with datum references and positional tolerance callouts.
- Designed around CO_2 laser-cut plywood panels, FDM inserts, fasteners, sanding/deburring, and actual assembly access.
- Validated prototype fit-up using calipers, fastener checks, squareness checks, lid closure, and latch/hinge alignment.

FIRST Robotics Hardware

Mechanism iteration, controls-aware CAD, manufacturing, match-side debugging

INTO THE DEEP robot CAD and subsystem packaging.



High-iteration intake mechanism CAD.

SNAPSHOT

ROLE	Hardware Lead for FTC #13216; mechanical design + TeleOp software
SCOPE	Drivetrain, intake, transfer, arm/lift, end effector, hang
WORK	Hundreds of CAD parts; printed, laser-cut, CNC, and shop-fabricated hardware
TOOLS	SolidWorks, Java, FTC SDK, PID, mecanum kinematics, FEA

ENGINEERING WORK

- Iterated intakes and scoring mechanisms around game-piece behavior, reliability, driver feel, and service access.
- Wrote Java TeleOp code for field-centric mecanum drive, mechanism sequencing, PID loops, and driver feedback.
- Competition robots had minimal mechanical failures and advanced through qualification into elimination rounds.

Robotics Hardware Capability Summary

Actuation + Mechanisms

- Compact Wolfrom compound planetary layout
- Bearing/output stackups and axial packaging
- Gear mesh, fastener access, assembly sequence
- Controls-aware mechanism design from FTC work

Electromechanical Systems

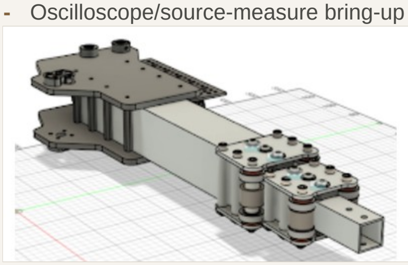
- 588 V accumulator and HV electronics attic packaging
- Board, connector, fan, service, and enclosure layout
- Mechanical/electrical interface tradeoffs
- Serviceability and inspection-aware packaging

PCB + Lab Hardware

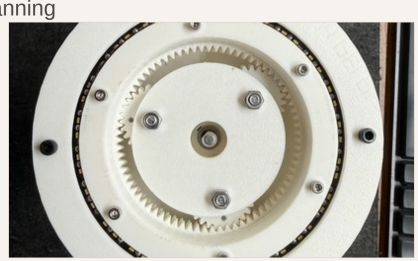
- KiCad schematic/layout workflow
- LTspice timing/switching checks
- GaN switching and trigger/sync paths
- Oscilloscope/source-measure bring-up planning

Manufacturing + Validation

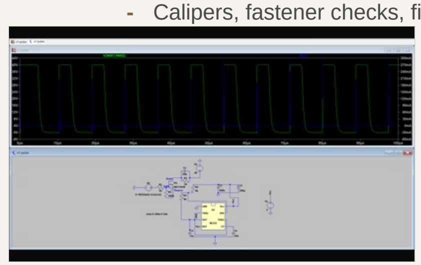
- Production-style drawings and datum strategy
- GD&T principles based on ASME Y14.5-2018
- Laser/FDM prototype fabrication constraints
- Calipers, fastener checks, fit-up validation



Linkage/end-effector packaging



Actuator internal stackup



Simulation-informed pulse workflow

Links

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Optional systems on the website include PLDTracker and Leitmotif; this PDF keeps the spotlight on hardware, CAD, PCBs, manufacturing, and validation.